

# E-Commerce Order Management

## RAG Test Knowledge Base

This document is intentionally self-contained and can be uploaded directly into a RAG application. The evaluation questions are designed to be answered using this document only.

### 1. Order Lifecycle

The order lifecycle begins when a customer places an order. A newly created order has the status **Pending**. After payment is successfully authorized, the order moves to **Confirmed**. When warehouse staff begin preparing the items, the status becomes **Processing**. After the package is handed to the carrier, the status becomes **Shipped**. Once the carrier confirms delivery, the status becomes **Delivered**.

### 2. Order Cancellation

Customers can cancel an order while its status is **Pending** or **Confirmed**. Cancellation is not allowed once an order reaches **Processing**, **Shipped**, or **Delivered**. If a customer cancels a paid order before processing begins, the payment is automatically marked for refund. Refund processing normally takes 3–5 business days after the refund is initiated.

### 3. Payment Statuses

The payment service uses the following statuses: **Open**, **Authorized**, **Captured**, **Failed**, **Refund Pending**, and **Refunded**. An Authorized payment has been approved by the payment provider but the money has not yet been captured. A Captured payment means the payment has been completed and the merchant has received the funds. A Failed payment does not create a successful payment transaction and can normally be retried.

### 4. Shipping

Standard shipping normally takes 3–5 business days after an order is shipped. Express shipping normally takes 1–2 business days after shipment. The shipping carrier can be identified from the tracking number prefix. Tracking numbers beginning with **EX** indicate Express shipments, while tracking numbers beginning with **ST** indicate Standard shipments.

### 5. Refund Rules

A customer can request a refund for a Delivered order within 30 calendar days of delivery. Items must be unused and returned with their original packaging. Once the returned item passes inspection, the refund is initiated. Refunds are sent to the original payment method. Shipping fees are refundable only when the return is caused by an incorrect item being shipped or an item arriving damaged.

### 6. Customer Support Priority

Support tickets are assigned one of three priorities. **High** priority is used for payment failures on an order that cannot be completed, duplicate charges, or orders marked Delivered but not received. **Medium** priority is used for delayed shipments and refund requests that have exceeded the normal processing period. **Low** priority is used for general product questions, address questions before shipment, and other informational requests.

### 7. Inventory Reservations

When an order is created, the requested quantity is reserved for 30 minutes while payment is being completed. If payment is not successfully authorized within 30 minutes, the inventory reservation expires and the quantity becomes available to other customers. Confirmed orders keep their inventory reservation until fulfillment.

## 8. Important Order Example

Order **ORD-78421** belongs to customer **CUST-2048**. It contains two units of the product **Wireless Keyboard K2** at \$45 per unit. The order subtotal is therefore \$90. The order uses Express shipping with a \$12 shipping fee, giving a total before tax of **\$102**. The payment transaction is **PAY-99182**. The order tracking number is **EX8472910**, which indicates Express shipping.

## 9. Another Order Example

Order **ORD-78435** belongs to customer **CUST-2077**. It contains one **USB-C Dock D4** priced at \$80. The order uses Standard shipping with a \$7 shipping fee. Its total before tax is **\$87**. The payment transaction is **PAY-99201**. Its tracking number is **ST5521804**, which indicates Standard shipping.

## 10. System Limits

The order search API returns a maximum of 50 orders per request. When more than 50 orders match a search, the client must use pagination. The default page size is 20 orders. The maximum supported page size is 50. Search results are sorted by order creation time in descending order by default.

## 11. Address Changes

A customer can change the shipping address while an order is **Pending** or **Confirmed**. Once the order reaches **Processing**, the shipping address can no longer be changed through the customer portal. Customers who need an address correction after processing begins must contact support.

## 12. Security

Customer support agents must never request or store a customer's full payment card number or card security code. For payment troubleshooting, agents should use the payment transaction ID and the last four digits of the card when available. API logs must mask authentication tokens and other credentials.